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SECTION B

*Answer **all** questions in the spaces provided.*

8. *(a)* Chloroethanoic acid is produced from ethanoic acid by passing chlorine gas into it under suitable conditions.

$$\underset{M_r\ 60.0}{CH_3COOH} + Cl_2 \longrightarrow \underset{M_r\ 94.5}{ClCH_2COOH} + HCl$$

(i) This reaction involves radicals. In the first stage chlorine radicals are formed. They then attack the ethanoic acid molecule producing a molecule of hydrogen chloride and a new radical.

Suggest the formula of the new radical formed from the acid during this reaction. [1]

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(ii) I. In an experiment 89.0 g of ethanoic acid reacted with chlorine.

Calculate the increase in mass that occurs if chloroethanoic acid is the only organic product. [2]

Increase in mass = ... g

II. Chloroethanoic acid reacts with ammonia producing aminoethanoic acid.

$$CH_3COOH \longrightarrow ClCH_2COOH \longrightarrow CH_2(NH_2)COOH$$

In the experiment described in I, 89.0 g of ethanoic acid then gave 49.2 g of aminoethanoic acid (M_r 75.0).

Calculate the percentage yield of aminoethanoic acid. Give your answer to an **appropriate** number of significant figures. [2]

Percentage yield = ... %

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III. Aminoethanoic acid is isolated from the solution of the products by adding methanol to the liquid. The acid is precipitated as a white solid.

Suggest why aminoethanoic acid is insoluble in methanol. [2]

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(b) Explain why an aqueous solution of aminoethanoic acid does not affect the plane of plane polarised light. [1]

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(c) Draw the structure of the dipeptide formed from two molecules of aminoethanoic acid. [1]

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(d) Aminoethanoic acid reacts with nitric(III) acid to produce nitrogen gas.

$$CH_2(NH_2)COOH \quad + \quad HNO_2 \longrightarrow CH_2(OH)COOH \quad + \quad N_2 \quad + \quad H_2O$$

Calculate the maximum volume of nitrogen gas produced at 100 °C and 98 kPa pressure when 0.300 mol of aminoethanoic acid reacts in this way. Give your answer in dm^3. [3]

Volume produced = ... dm^3

(e) Hydroxyethanoic acid, $CH_2(OH)COOH$, can also be prepared by reacting chloroethanoic acid with aqueous sodium hydroxide.

Explain why the reaction mixture needs to be acidified to produce hydroxyethanoic acid. [1]

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